04 Sample Size - Riley et al. (2019) Power Calculation

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0.1 Introduction

Riley et al. (2019) provide minimum sample size criteria for prediction models: (i) shrinkage ≥ 0.9, (ii) Nagelkerke R^2 optimism ≤ 0.05, and (iii) adequate event-rate precision.

The criteria assume independent observations, and person-hours are not independent: they are nested within ICU stays at roughly 45 correlated rows per stay, one stay per patient. The adequacy assessment is therefore parameterised on the **stay**: the outcome proportion supplied to `pmsampsize` is the fraction of training *stays* labelled septic, and the resulting minimum is compared against the number of training stays.

The per-person-hour parameterisation is also computed and retained, because it is the quantity that governs the precision of an hourly hazard estimate conditional on the clustering, and because reporting only one of the two invites the reader to compare a minimum derived at one prevalence against a count at the other. It is **not** an effective sample size and must never be compared against the 2.3M person-hour row count as though it were. The stay-level figure is the one the thesis reports as the adequacy verdict.

The calculation is repeated per label variant. Anticipated C-statistic = 0.846 (Moor et al., 2023).

0.2 Label-Variance Recap

Cohort and event counts from the label-variance table (produced in 02) are the inputs.

Table 1: Label-variance table (from 02_cohort_labels.Rmd)

variant	name	n_stays	n_sepsis_onsets	pct_sepsis	n_non_sepsis	n_deaths_no_sep	m
A	Narrow	65078	6252	9.6	58826	5934	
B	Seymour-Standard	65078	6447	9.9	58631	5892	
C	Liberal	65078	12581	19.3	52497	4956	

0.3 Sample Size per Variant

For each variant, the training-set person-hour table is loaded and `pmsampsize` called with the anticipated C-statistic of 0.846. Events-per-variable (EPV) is reported as a secondary reference.

```
##
## --- Variant A (Narrow) ---
##   Training person-hours: 2,304,923
##   Training stays:        52,946
##   Onset events (train):  4,647
##   Per-hour event rate:   0.2016%
##   Stay-level sepsis prevalence: 8.7769%
##   [PRIMARY] Minimum independent observations (Riley, stay-level): 1,729
##             Available independent units (training stays):        52,946
##             Sufficient: YES (margin 30.62x)
##   [reference] Minimum at the per-hour prevalence: 60,553
##              Rows are clustered within stays, so the person-hour count
##              does not supply that many independent observations.
##   Events-per-variable (EPV): 160.2
##
## --- Variant B (Seymour-Standard) ---
##   Training person-hours: 2,301,842
##   Training stays:        52,946
##   Onset events (train):  4,847
##   Per-hour event rate:   0.2106%
##   Stay-level sepsis prevalence: 9.1546%
##   [PRIMARY] Minimum independent observations (Riley, stay-level): 1,666
##             Available independent units (training stays):        52,946
##             Sufficient: YES (margin 31.78x)
##   [reference] Minimum at the per-hour prevalence: 59,174
##              Rows are clustered within stays, so the person-hour count
##              does not supply that many independent observations.
##   Events-per-variable (EPV): 167.1
##
## --- Variant C (Liberal) ---
##   Training person-hours: 2,120,376
##   Training stays:        52,946
##   Onset events (train):  9,403
##   Per-hour event rate:   0.4435%
##   Stay-level sepsis prevalence: 17.7596%
##   [PRIMARY] Minimum independent observations (Riley, stay-level): 991
##             Available independent units (training stays):        52,946
##             Sufficient: YES (margin 53.43x)
##   [reference] Minimum at the per-hour prevalence: 27,920
##              Rows are clustered within stays, so the person-hour count
##              does not supply that many independent observations.
##   Events-per-variable (EPV): 324.2
```

0.4 Summary and Interpretation

The table below reports the adequacy verdict on the independent unit: the minimum number of independent observations required at each variant's stay-level prevalence, against the number of training stays available. The person-hour columns are the size of the design matrix and are reported for completeness only.

```
##
## Saved: 04_sample_size.parquet
```